연성범 (Seongbeom Yeon)

Department of Materials Science and Engineering
Korea Advanced Institute of Science and Technology (KAIST)
291 Daehak-ro, Yuseong-gu, Daejeon 34141, Korea
ORCID: 0000-0002-5436-8674
Homepage: https://yesbeom.github.io

Education

Sep. 2021 − Present	**Integrated M.S. and Ph.D., Materials Science and Engineering** KAIST, Daejeon, Korea (Advisor: Professor Himchan Cho)
Sep. 2014 − Feb. 2021 (Military Service: Feb. 2016 – Nov. 2017)	**Integrated B.S. and M.S., Chemistry** Chungbuk National University, Cheongju, Korea ▪ **Thesis title**: (An) Experimental and Theoretical Study for Synthesis of new n-type Zintl Phase Compounds : The $Ca_{5-x-y}Yb_xRE_yAl_2Sb_6$ (RE =Pr, Nd, Sm, Gd, Eu; $0 \le x \le 4.82(4)$; $0 \le y \le 0.44(1)$), Advisor: Prof. Tae-Soo You

Research Highlightes

† (equally contributed 1st author)

1. Changhyun Joo†, **Seongbeom Yeon†**, Jordi Llusar, Shehla Gul, Chanho Park, Junhui Lee, Jaeyeong Ha, Jaedong Jang, In Hye Kwak, Hyung Joong Yun, Sang Kyu Kim, Yong-Hoon Cho, Ivan Infante*, Himchan Cho*, "Overcoming the Luminescence Effiency Limitations of InP Magic-Sized Clusters"
   Journal of the American Chemical Society **2025**, 52, 48046–48059 (SCI, IF = 16.6)

2. **Seongbeom Yeon**, Yoseph Kim, Abdessamad El Adel, Jaeyeong Ha, Seongkyu Maeng, Youngjo Kim, Ivan Infante*, Himchan Cho*, "Olefin Ligand Metathesis for Colloidal Emissive Nanocrystals with Enhanced Stability and Photosensitivity"
   Angewandte Chemie International Edition **2025**, e202514802 (SCI, IF = 16.9)
   (Selected as a **Hot paper**)

3. Jaehwan Lee†, **Seongbeom Yeon†**, Bonwoo Koo, Bokyeong Sohn, Sun Jae Park, Cheoljae Kim, Himchan Cho*, "Photocleavable Ligand-Induced Direct Photolithography of InP-Based Quantum Dot"
   ACS Energy Letters **2025**, 10, 94–101 (SCI, IF = 19.5)
   (Selected as a **supplementary cover**)

4. Jaeyeong Ha†, **Seongbeom Yeon†**, Jaehwan Lee, Hyungdoh Lee, Himchan Cho*, "Revealing the Role of Organic Ligands in Deep-Blue-Emitting Colloidal Europium Bromide Perovskite Nanocrystals"
   ***ACS Nano* 2024**, 18, 31891-31902 (SCI, IF = 15.881)
   (Selected as a **supplementary cover**)

5. **Seongbeom Yeon**, Seungeun Shin, Hongil Jo, Nguyen Le Thi, Dohyun Moon, Dong-Hyun Kim, Kang Min Ok, Tae-Soo You*, "p-Type to n-Type Conversion through the "Bypass" Phase Transition in the Zintl-Phase Thermoelectric Materials"
   ***Chem. Mater.* 2021**, 33, 6761–6773 (SCI, IF = 10.508)

Selected Honors & Awards

† Joint winners (equally contributed 1st author)

Jun. 2026	**GCIM 2026 Best Poster Presentation Award** *International Conference* Materials Research Society of Korea, Jeju, Korea
May. 2026	**QD2026 Best Poster Presentation Award†** *International Conference* The Korean Information Display Society, Busan, Korea
Jul. 2025	**Excellent Paper Presentation Award†** Materials Research Society of Korea, Gangneung, Korea
Apr. 2024	**MRS Spring 2024 Best Poster Award†** *International Conference* Materials Research Society (MRS) Spring Meeting & Exhibit, Seattle, Washington, USA
Nov. 2023	**Excellent Paper Presentation Award†** Korean Society of Industrial and Engineering Chemistry, Gwangju, Korea
Nov. 2022	**Excellent Paper Presentation Award** Korean Society of Industrial and Engineering Chemistry, Daejeon, Korea
Nov. 2021	**Best Paper Award** Korean Chemical Society, Seoul, Korea
May. 2019	**Excellent Oral Presentation Award** College of Natural Sciences, Chungbuk National University, Cheongju, Korea

Scholarships

Sep. 2021 – Present	**Full Scholarship** KAIST, Daejeon, Korea
Sep. 2019 – Aug. 2020	**Full Scholarship** Chungbuk National University, Cheongju, Korea
Mar. 2016 – Aug. 2019	**National Scholarship for Science and Engineering** Ministry of Science and ICT, Sejong, Korea
Sep. 2015	**Scholarship for Knowledge Mentoring** Chungbuk National University, Cheongju, Korea
Mar. 2015 – Feb. 2016	**Scholarship for Outstanding Student** Chungbuk National University, Cheongju, Korea
Mar. 2014 – Feb. 2015	**National Scholarship** Korea Student Aid Foundation, Seoul, Korea

Professional Activities

Sep. 2022	***Organization Staff*** **The Global Networking Workshop for New Researchers of Nanoeletronic Materials** Online Meeting
Fall 2021	***Teaching Assistant*** **General Chemistry I** KAIST, Daejeon, Korea
Nov. 2020	***Organization Staff*** **The 2020 Central Region Inorgnic Chemistry Symposium** Chungbuk National University, Cheongju, Korea
Fall 2020	***Teaching Assistant*** **Inorganic Chemistry I** Chungbuk National University, Cheongju, Korea
Spring 2020	***Teaching Assistant*** **General Chemistry I** Chungbuk National University, Cheongju, Korea
Jan. 2020	***Organization Staff*** **The 2020 Winter Inorganic Solid-State Chemistry Workshop** Gwangju-si, Gyeonggi-do, Korea
Fall 2019	***Teaching Assistant*** **General Chemistry II** Chungbuk National University, Cheongju, Korea

Jan. 2019 ***Organization Staff***
The 2019 Winter Inorganic Solid-State Chemistry Workshop
Pyeongchang, Korea

Full Publications

† (equally contributed 1st author)

Under Review or In preparation

1. Bokyeong Sohn, **Seongbeom Yeon**, Himchan Cho*, " Materials Design Constraints and Synthetic Strategies for Longer Wavelength Infrared Quantum Dots", *In preparation*
2. Bokyeong Sohn, **Seongbeom Yeon**, Hwa Seob Choi, Jaehwan Lee, Jae Jun Lee, Jaedong Jang, Hyungdoh Lee, Himchan Cho*, "Unraveling the Effect of Metal Halide Complex on the Surface Chemistry of InP Quantum Dots", *Under Review*

Published

3. Changhyun Joo†, **Seongbeom Yeon†**, Jordi Llusar, Shehla Gul, Chanho Park, Junhui Lee, Jaeyeong Ha, Jaedong Jang, In Hye Kwak, Hyung Joong Yun, Sang Kyu Kim, Yong-Hoon Cho, Ivan Infante*, Himchan Cho*, "Overcoming the Luminescence Effiency Limitations of InP Magic-Sized Clusters"
Journal of the American Chemical Society **2025**, 52, 48046–48059 (SCI, IF = 16.6)

4. Jaehwan Lee†, Yonghan Jo†, **Seongbeom Yeon**, Jinha Jang, Jaeyeong Ha, Bokyeong Sohn, Hyunjin Cho, Gui-Min Kim, Doh C. Lee, Chan Beum Park*, and Himchan Cho*, "All-Inorganic InP/ZnSe/ZnS Quantum Dots with Chalcogenidometallate Ligands for Biosolar Catalysis"
Advanced Functional Materials 2025, e11671 (SCI, IF = 19)

5. **Seongbeom Yeon**, Yoseph Kim, Abdessamad El Adel, Jaeyeong Ha, Seongkyu Maeng, Youngjo Kim, Ivan Infante*, Himchan Cho*, "Olefin Ligand Metathesis for Colloidal Emissive Nanocrystals with Enhanced Stability and Photosensitivity"
Angewandte Chemie International Edition **2025**, e202514802 (SCI, IF = 16.9)
(Selected as a **Hot paper**)

6. Jaehwan Lee†, **Seongbeom Yeon†**, Bonwoo Koo, Bokyeong Sohn, Sun Jae Park, Cheoljae Kim, Himchan Cho*, "Photocleavable Ligand-Induced Direct Photolithography of InP-Based Quantum Dot"
ACS Energy Letters **2025**, 10, 94–101 (SCI, IF = 19.5)
(Selected as a **supplementary cover**)

7. Jaeyeong Ha†, **<u>Seongbeom Yeon†</u>**, Jaehwan Lee, Hyungdoh Lee, Himchan Cho*, "Revealing the Role of Organic Ligands in Deep-Blue-Emitting Colloidal Europium Bromide Perovskite Nanocrystals"
   ACS Nano **2024**, 18, 31891-31902 (SCI, IF = 15.881)
   (Selected as a **supplementary cover**)

8. Junsu Lee, Naeun Seo, **<u>Seongbeom Yeon</u>**, Tae-Soo You*, "Novel n-type Zintl Phase Thermoelectric Materials: The $Ca_{5-x}RE_xAl_2Sb_6$ (RE = Pr, Sm)"
   Bull. Kor. Chem. Soc. **2023**, 44, 705–712 (SCI)

9. Sujin Kim, Junphil Hwang, Tae-Soo You, **<u>Seongbeom Yeon</u>**, Jungwon Kim, Byung-Kyu Yu, Mi-Kyung Han, Minju Lee, Somnath Acharya, Jiyong Kim, Woochul Kim, and Sung-Jin Kim*, "Enhanced Thermoelectric Performance by Resonant Doping and Embedded Magnetic Impurity"
   Phys. Rev. Applied. **2022**, 19, 014034. (SCI, IF = 4.7)

10. Yeongjin Hong, **<u>Seongbeom Yeon</u>**, Philip Yox, Zhao Yunxiu, Dohyun Moon, KangMin Ok, Dong-Hyun Kim, Kirill Kovnir, Gordon J. Miller, and Tae-Soo You*, "Role of Eu-Doping on the electron Transport Behavior in the Zintl Thermoelectric $Ca_{5-x-y}Yb_xEu_yAl_2Sb_6$ System"
    Chem. Mater. **2022**, 34, 9903–9914. (SCI, IF = 8.6)
    (Selected as a **front cover**)

11. Yeongjin Hong†, **<u>Seongbeom Yeon</u>**†, Jiwon Jeong, Dohyun Moon, and Tae-Soo You*, "Rare-Earth Metal Doped Zintl Phase Thermoelectric Materials: the $Yb_{5-x}RE_xAl_2Sb_6$ (RE = Pr, Nd, Sm) System"
    Bull. Kor. Chem. Soc. **2022**, 43, 1191–1199 (SCI, Selected as a **front cover**)

12. Jee Yoon Chung, **<u>Seongbeom Yeon</u>**, Hongsun Ryu, Tae-Soo You*, Joon I. Jang*, Kang Min Ok*, "Nonlinear Optical Properties of A New Polar Bismuth Tellurium Oxide Fluoride $Bi_3F(TeO_3)(TeO_2F_2)_3$"
    J. Alloys Compd. **2022**, *895*, 162603. (SCI, IF = 6.371)

13. **<u>Seongbeom Yeon</u>**, Seungeun Shin, Hongil Jo, Nguyen Le Thi, Dohyun Moon, Dong-Hyun Kim, Kang Min Ok, Tae-Soo You*, "p-Type to n-Type Conversion through the "Bypass" Phase Transition in the Zintl-Phase Thermoelectric Materials"
    Chem. Mater. **2021**, 33, 6761–6773 (SCI, IF = 10.508)

14. Junsu Lee, Yeongjin Hong, Seongbeom Yeon, Dohyun Moon, Tae-Soo You*, "Effect of Cationic and Anionic Doping in the Quinary Zintl Phase Thermoelectric Material $Ca_{5-x}Yb_xAl_{2-y}In_ySb_{6-z}Sn_z$ System"
    Bull. Kor. Chem. Soc. 2021, 42, 563–566. (SCI)

15. Junsu Lee, Seongbeom Yeon, Tae-Soo You*, "Development of Thermoelectric Materials Converting Various Thermal Energy into Electricity"
Chemworld 2020, 7, 25–34. / Domestic academic journals

16. Jee Yoon Chung, Hongil Jo, Seongbeom Yeon, Hye Ryung Byun, Tae-Soo You, Joon I. Jang, Kang Min Ok*, "$Bi_3(SeO_3)_3(Se_2O_5)F$: A Polar Bismuth Selenite Fluoride with Polyhedra of Highly Distortive Lone Pair Cations and Strong Second-Harmonic Generation Response"
Chem. Mater. 2020, 32, 7318–7326. (SCI, IF = 9.811)
(Selected as a **supplementary cover**)

Patents

1. **Domestic granted patents (국내 특허 등록)**
   1) 유태수, 연성범, "희토류 금속 양이온, 칼슘 양이온, 및 이터븀 양이온이 포함된 오성분계 진틀상 화합물 및 이의 제조방법 (Rare-earth Metal Cation, Calcium Cation, And Ytterbium Cation Mixed Quinary Zintl Phase Compound And Manufacturing Method Thereof)", Korea,
   Application No.: 10-2021-0078003 (2021.06.16),
   Grant: 1025235450000 (2023.04.14)
   2) 유태수, 연성범, "유로퓸이 도입된 진틀화합물 및 이의 제조방법 (Eu-Doping Zintl Compound And Manufacturing Method Thereof)", Korea,
   Application No.: 10-2022-0103603 (2022.08.18),
   Grant: 1024922020000 (2023.01.19)

2. **Domestic patent applications (국내 특허 출원)**
   1) 조힘찬, 연성범, "올레핀 복분해 반응을 이용한 무기 나노입자의 개질 방법, 무기 나노입자를 포함하는 감광성 조성물 및 이를 이용한 패턴 형성 방법", Korea,
   Application No.: 10-2025-0026528 (2025.02.28)
   2) 조힘찬, 이재환, 연성범, "광절단 가능한 광민감성 리간드 기반의 양자점 패터닝 기술 (Quantum dots containing light-activated ligands, a quantum dot patterning method using the same, and an electroluminescent device)", Korea,
   Application No.: 10-2024-0157970 (2024.11.08)
   3) 조힘찬, 연성범, "무기 다중 금속 칼코제나이드 복합체 리간드를 갖는 나노입자, 이를 포함하는 광감성 조성물 및 이를 이용한 패턴 형성 방법 (NANO-PARTICLE HAVING INORGANIC MULTINARY METAL CHALCOGENIDE COMPLEX LIGAND, PHOTOSENSITIVE COMPOSITION INCLUDING THE SAME AND METHOD FOR FORMING PATTERN USING THE SAME)", Korea,

Application No.: 10-2024-0140399 (2024.10.15), Priority Application No.: 10-2023-0139441 (2023.10.18)

4) 조힘찬, 이재환, 연성범, "광민감성 무기 리간드 및 나노입자를 활용한 EUV 용 포토레지스트 조성물 및 이를 이용한 패턴 형성 방법 (PHOTORESIST COMPOSITION USING PHOTO-SENSITIVE INORGANIC LIGAND AND INORGANIC NANO-PARTICLES FOR EUV AND METHOD FOR FORMING PATTERN USING THE SAME)", Korea,
Application No.: 10-2024-0138998 (2024.10.11), Priority Application No.: 10-2023-0159222 (2023.11.16)
5) 조힘찬, 이재환, 이승우, 김경민, 연성범, 이민구, 김근영, "반도체 소자, 이를 포함하는 이미지 처리 장치, 및 이미지 처리 장치의 동작 방법 (SEMICONDUCTOR DEVICE, IMAGE PROCESSING APPARATUS INCLUDING THE SAME, AND OPERATING METHOD OF IMAGE PROCESSING DEVICE)", Korea,
Application No.: 10-2024-0101667 (2024.07.31)
6) 조힘찬, 하재영, 연성범, "유로퓸 기반 발광 나노결정 및 용액 합성을 통한 유로퓸 기반 발광 나노결정의 제조 방법 (EUROPIUM-BASED LIGHT-EMITTING NANOCRYSTALS AND METHOD FOR FABRICATING EUROPIUM-BASED LIGHT-EMITTING NANOCRYSTALS THROUGH COLLOIDAL SYNTHESIS)", Korea,
Application No.: 10-2024-0041172 (2024.03.26)

Press